Nepal PreTST 2024 Solutions

Prakrit Gajurel

28 June 2026

§1 Problems

Problem 1.1. Nirajan is trapped in a magical dungeon. He has infinitely many magical cards with arbitrary MPs(Mana Points) which is always an integer $\mathbb{Z}$. To escape, he must give the dungeon keeper some magical cards whose MPs add up to an integer with at least 2024 divisors. Can Nirajan always escape?

Proof. □

Problem 1.2. Let, $S = \sum_{i=1}^{k} {n_i}^2$. Prove that for $n_i \in \mathbb{R}^+$

$$\sum_{i=1}^{k} \frac{n_i}{S - n_i^2} \geq \frac{4}{n_1 + n_2 + \cdots + n_k}$$

Proof. □

Problem 1.3. Let ABC be an acute triangle and H be its orthocenter. Let E be the foot of the altitude from C to AB, F be the foot of the altitude from B to AC. Let $G \neq H$ be the intersection of the circles (AEF) and (BHC). Prove that AG bisects BC.

Proof. □

Problem 1.4. Find all integer/s n such that $\frac{5^n - 1}{3}$ is a prime or a perfect square of an integer.

Proof. □